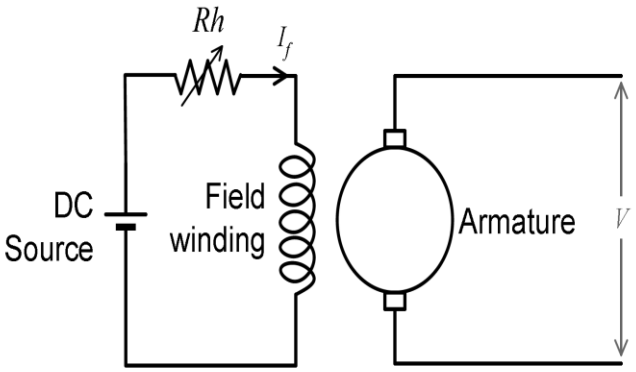
Module 1
DC Machines

- **DC shunt generator - no-load (open circuit) characteristics**
- **critical field resistance, critical speed, voltage buildup -**

Open circuit characteristics of shunt generator

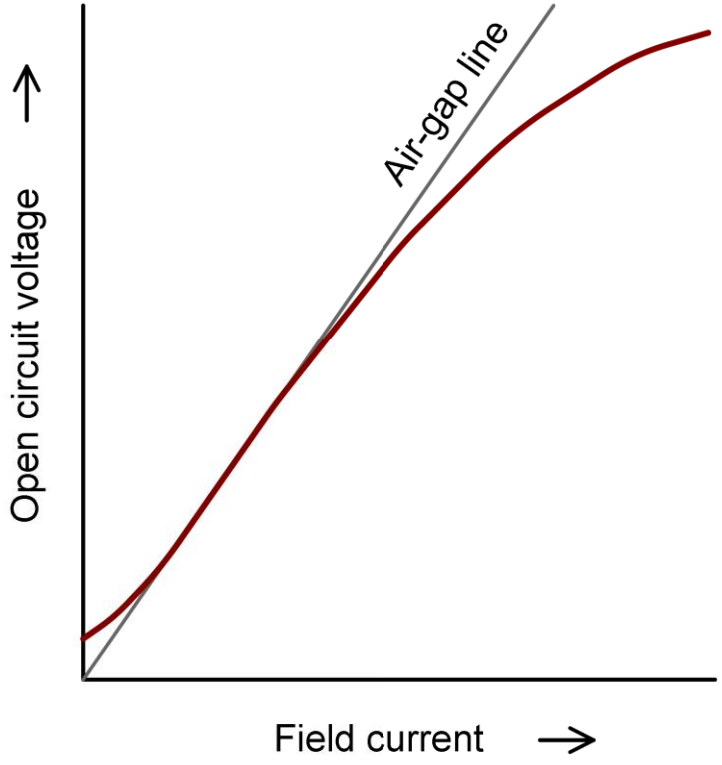
How is OCC plotted?

- The field winding is excited from a DC source and the field current is varied in steps by adjusting the rheostat connected in series
- The generated voltage across the armature terminals is recorded for each value of field current and the curve is plotted
- OCC is the plot of terminal voltage of the DC generator against field current and without any load connected to it



Open circuit characteristics of shunt generator

- The characteristics make a curve at higher values of field current due to the magnetic saturation of the core material.
- There is a small voltage induced even when field current is zero. This is due to the residual magnetism in the core
- The tangent drawn at the lower part of the OCC curve is called air-gap line
- Airgap line represents the mmf required to overcome the reluctance of the airgap



Terms associated with OCC

- **Residual voltage**

  Small voltage induced in the armature when field current is zero

- **Critical resistance**

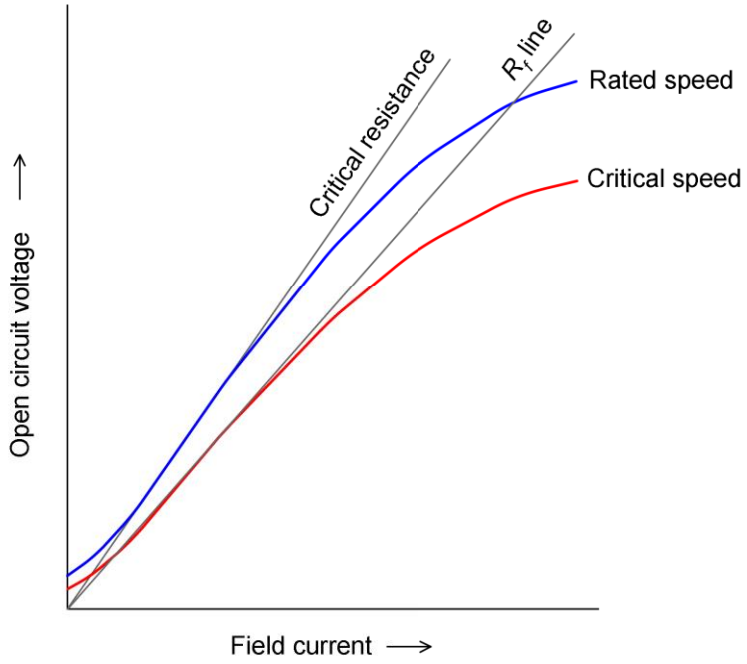
  The resistance of the field circuit above which the shunt generator will fail to excite at rated speed operation

- **R_f line**

  Line representing the resistance of the field winding without any external resistance

- **Critical speed**

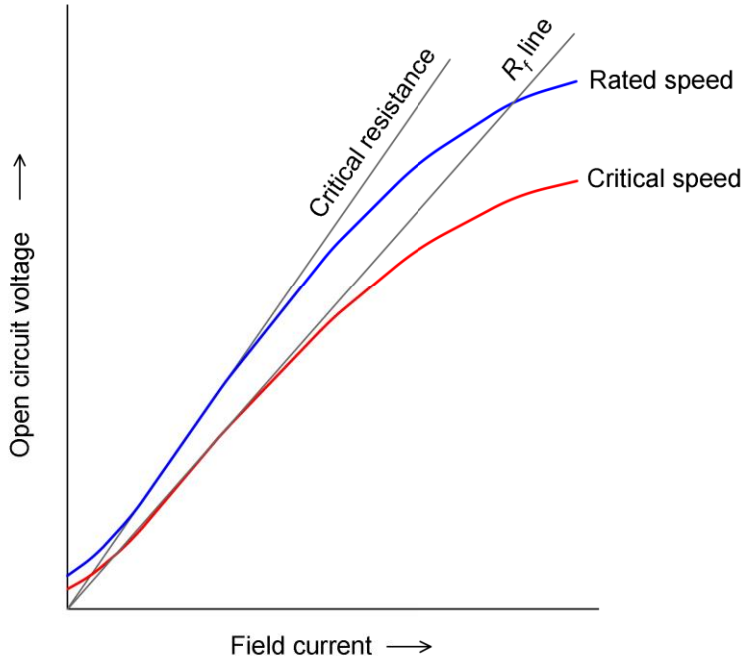
  Speed below which a dc shunt machine will fail excite even at minimum resistance in field circuit



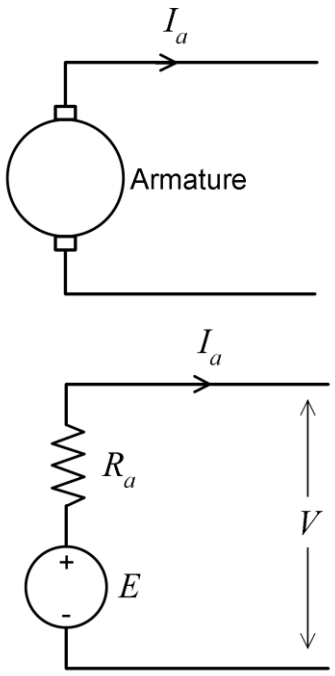
Conditions for voltage build up in a DC shunt generator

Essential conditions for voltage buildup in dc shunt generators

- Residual magnetism should be present in the pole core
- The connections of the armature and field windings should be in such a way that the voltage induced due to residual magnetism will further increase the flux in the pole
- The field circuit resistance should not be above the critical resistance
- The speed of the machine should be above the critical speed



Load Characteristics



The terminal voltage of the generator is given by

$$V = E - I_a R_a$$

where V = terminal voltage
E = induced voltage
I_a = armature current
R_a = armature resistance

For series and compound generators

$$V = E - I_a \left(R_a + R_s \right)$$

R_s = resistance of the series field winding

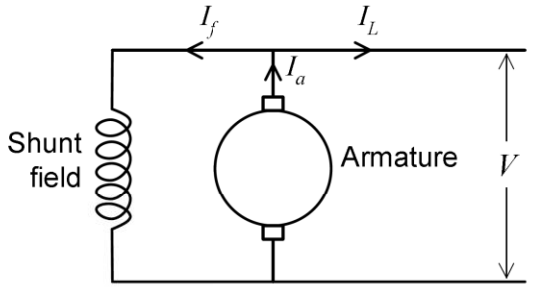
Example 2.1

A DC shunt generator has a terminal voltage of 220 V, field current of 0.5 A, and load current of 12 A. Find the induced voltage if the armature resistance is 0.2 Ω

Armature current, $I_a = I_L + I_f = 12 + 0.5 = 12.5$ A

We have $V = E - I_a R_a$

Then $E = V + I_a R_a = 220 + 12.5 \times 0.2 = \mathbf{222.5\ V}$



Example 2.2

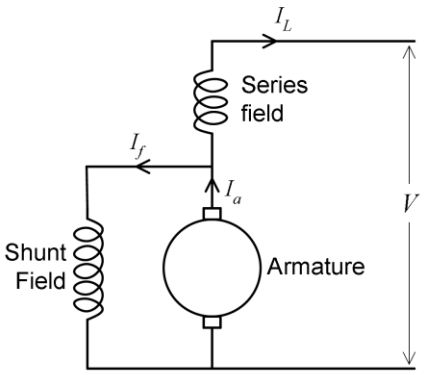
A compound wound dc generator with short shunt connection has an armature resistance of 0.15 Ω and a series field winding resistance of 0.1 Ω. The terminal voltage is 220 V when the machine delivers a current of 21A. If the shunt field current is 1 A, find the induced voltage.

Armature current, $I_a = 21 + 1 = 22$ A

Voltage drop in armature = $22 \times 0.15 = 3.3$ V

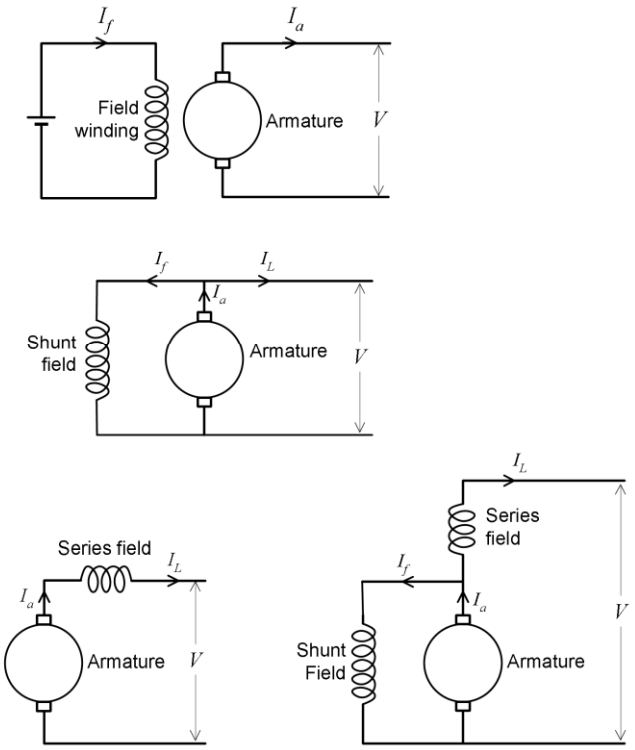
Voltage drop across series field winding = $21 \times 0.1 = 2.1$ V

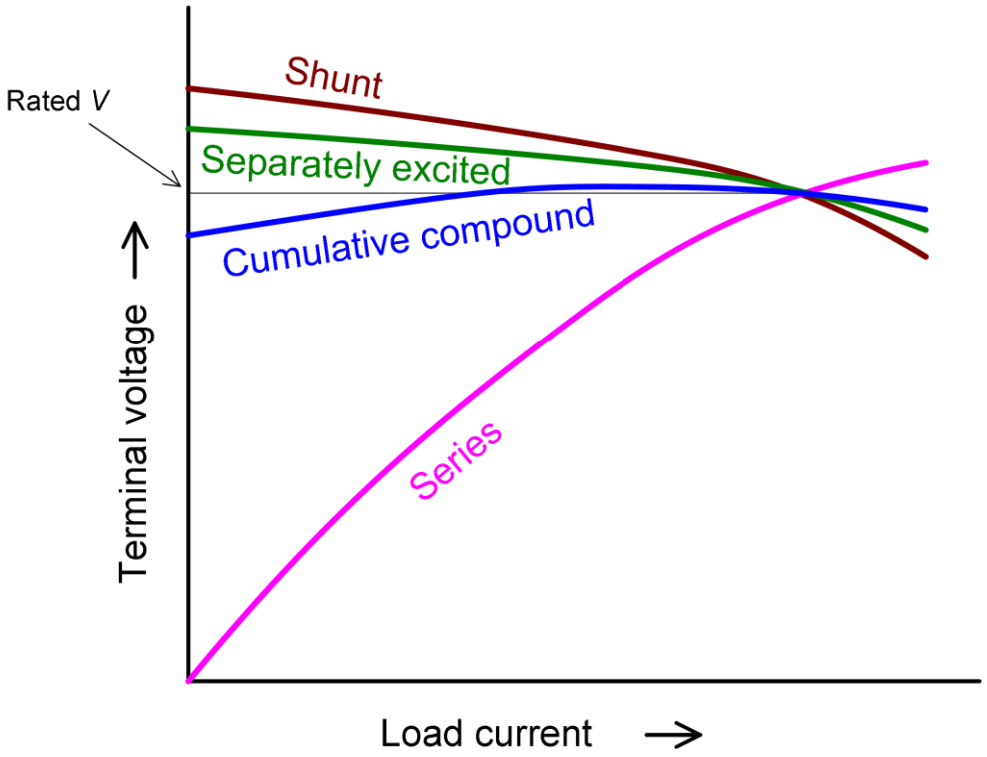
Induced voltage = $220 + 3.3 + 2.1 =$ **225.4 V**



$$E = V + I_a R_a + I_s R_s$$

Generator characteristics





Example 2.3

A dc shunt generator has the following characteristics when operating at its rated speed of 1500 rpm

I_f	0	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0
E	6	50	101	152	187	212	229	245	251

The field winding resistance is 325 ohms. Find the following

a) The voltage induced with no additional resistance in field circuit
b) Resistance to be added to generate open circuit voltage of 220V
c) Critical resistance
d) Critical speed

Steps:

- Plot the OCC corresponding to 1500 rpm based on the given data
- Plot the R_f line – find (a) from the graph
- Find I_f corresponding to 220 V from the graph – find field total field circuit resistance – find resistance to be added
- Plot tangent to OCC to get critical resistance. Also calculate critical speed from graph

a Draw the R_f line and from the point it cuts the OCC obtain the voltage as 240 V

b Field current corresponding to $220\text{V} = 0.55\ \text{A}$

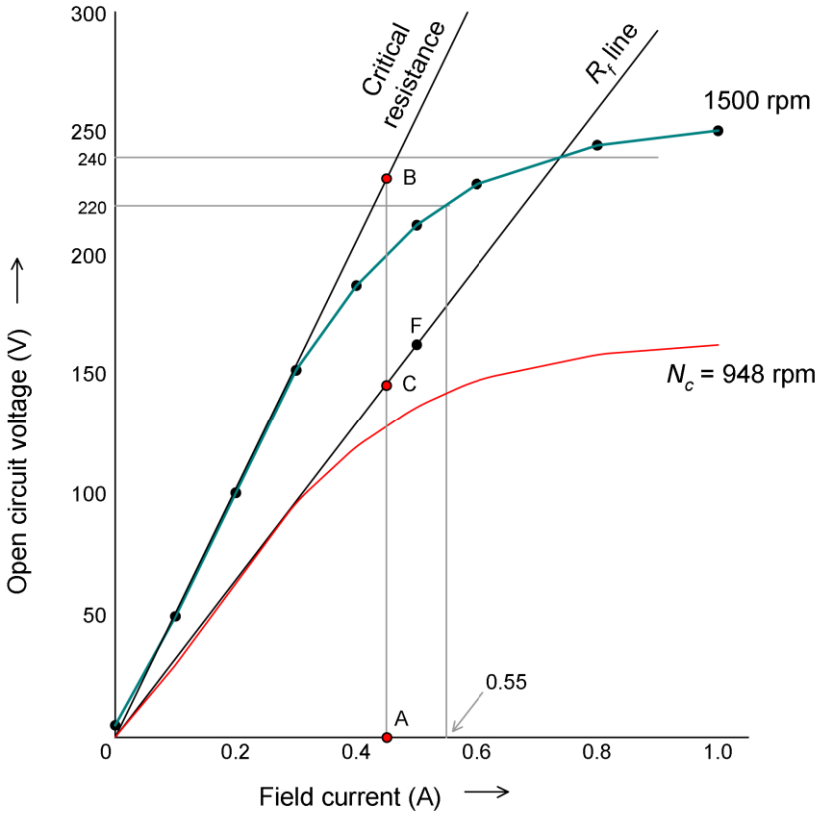
$$\text{Resistance of field circuit} = \frac{220}{0.55} = 400\ \Omega$$

$$\text{Additional resistance} = 400 - 325 = 75\ \Omega$$

c
$$\text{Critical resistance} = \frac{AB}{OA} = \frac{231}{0.45} = 513\ \Omega$$

d
$$\text{Critical speed} = \text{Rated speed} \times \frac{AC}{AB}$$

$$= 1500 \times \frac{146}{231} = 948\ \text{rpm}$$



Example 2.4

A long shunt DC compound generator delivers 25 kW power at 500 V. The resistance of armature, series and shunt winding are 0.05, 0.03 and 200 ohms respectively. Calculate generated emf and armature current. Assume 1 V per brush for contact drop.

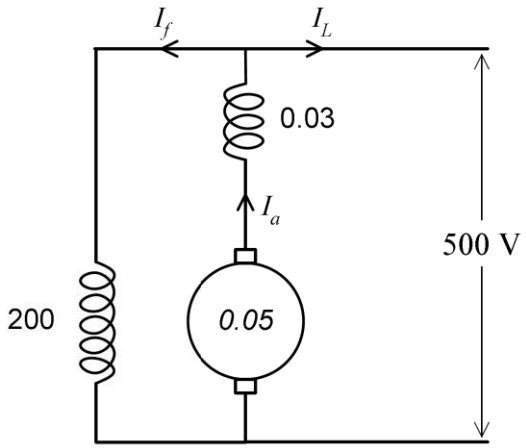
Load voltage, $V_L = 500$ V Power, $P = 25$ kW

Load current, $I_L = \frac{P}{V_L} = \frac{25 \times 10^3}{500} = 50$ A

Field current, $I_f = \frac{V_L}{R_f} = \frac{500}{200} = 2.5$ A

Armature current, $I_a = I_L + I_f = 50 + 2.5 = 52.5$ A

Generated voltage, $E = 500 + 52.5 \times (0.05 + 0.03) + 2 \times 1 = 506.2$ V



Example 2.5

Two DC shunt generators with induced emfs of 230V and 225V, armature resistance of 0.5 Ω and 0.4 Ω and field resistances of 100 Ω and 110 Ω respectively are in parallel with a terminal voltage of 220V. Calculate the power delivered by each generator.

Field current of the first generator,

$$I_{f1} = \frac{220}{100} = 2.2 \text{ A}$$

Field current of the second generator,

$$I_{f2} = \frac{220}{110} = 2 \text{ A}$$

Equivalent circuit:

